



Benjamin Meyer <ben@kenaiwatershed.org>

Follow-up on Mat-Su Lakes

5 messages

Benjamin Meyer <ben@kenaiwatershed.org>

Mon, Jun 8, 2026 at 10:26 AM

To: matt.s.mcmillan@gmail.com

Hi Matthew,

Greetings from Wasilla. LinkedIn tells me you are in Colorado these days, congrats on the move! My friend Galen Hecht also works for Resource Data in the Anchorage office.

I was just taking a look again at your 2016 thesis. I am wondering if you can share the original water quality data from this work. I know I could find someone at the Mat-Su Borough to dig it out for me, but I thought asking you might be simpler. I thought I already had it somewhere from Melanie from years ago but I do not. If you have it and are able to share, I would be very grateful.

In my role at Kenai Watershed Forum, I've learned a lot about the value of community-based water quality monitoring ([see our Kenai River monitoring program](#)). Also, I've learned how easy it is for these kind of data to evaporate if left unattended. Right before the program was cut, Melanie had various ideas for properly archiving the data and making it more user-friendly. I have experience with this kind of thing now, and I would love to put it on my to-do list. If the results suggest it could be useful, perhaps even revive the monitoring effort in some other form.

Anyways, for now, my only plan was to see if you are able to share a copy of the data. If not, I'll reach out to the borough.

Thanks Matt. All the best --

Ben

----- Forwarded message -----

From: **Benjamin M** <fsbem4@gmail.com>

Date: Mon, Jun 8, 2026 at 10:12 AM

Subject: Fwd: Beaver Creek (Kenai area) follow up

To: <ben@kenaiwatershed.org>

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From: **Matt McMillan** <matt@greatlandtrust.org>

Date: Fri, Apr 24, 2020 at 12:11 PM

Subject: Re: Beaver Creek (Kenai area) follow up

To: Benjamin Meyer <bemeyer@alaska.edu>

Hey Ben!

Thanks for reaching out again and glad to hear you continue to make progress with your research. It was good to see your presentation last year to learn more about the fish in the Kenai watershed.

We aren't really looking to do any projects on the Kenai anymore, hoping to defer any conservation work to Kachemak Heritage Land Trust since they have more experience there. You may get in touch with them since I think they are

actively looking for mitigation projects to offset certain road development projects that will impact the Kenai River.

As for proactive conservation strategies, KHLT uses a decision support tool called the Kenai Decision Support Tool (<http://gis.audubon.org/kenai/>) developed by Audubon Alaska to help in their priorities. We use similar GIS prioritizations to help us target land for conservation. We work very closely with Mike Daigneault (Michael_Daigneault@fws.gov) on our strategies so if Suzanne hasn't spoken to him yet she might be interested.

Hope you are doing well!

Matthew S. McMillan

Stewardship Coordinator & GIS Analyst



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On Thu, Apr 23, 2020 at 4:23 PM Benjamin Meyer <bemeyer@alaska.edu> wrote:

Hi Matt,

Hope you are well. We met last fall at the Mat-Su salmon symposium. I wanted to get in touch with you because you mentioned GLT had done some exploration to look into habitat conservation in the Beaver Creek area near Kenai.

I've been in communication with Suzanne Nelson, an FWS biologist who is tasked with choosing an example watershed to examine feasibility of "proactive conservation" strategies. If you or GLT are still examining issues related to Beaver Creek at all she'd be interested to learn more.

As for my own research, I've made more sense of my data since last fall too and am happy to share what I know about the area as well :)

Be in touch and let me know if GLT is still considering Beaver Creek anywhere in its portfolio of interests.

Take care! Cheers -- Ben

--

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"Find your true weakness and surrender to it. Therein lies the path to true genius." -Moshe Feldenkrais



McMillan_Thesis_Final_April_20163.pdf
4094K

Matt McMillan <matt.s.mcmillan@gmail.com>
To: Benjamin Meyer <ben@kenaiwatershed.org>

Mon, Jun 8, 2026 at 2:51 PM

Ben,

Thanks for reaching out! Man, if I knew what I know now, I sure would have taken better care of my organization of that data! I'll take a look at my hard drive to see what I can share, I feel like I uploaded it to some location for MatSu Borough but I don't remember. I'll let you know what I can find sometime this week. Say hi to Galen for me and hope the summer weather in Kenai is treating you well!

Matt
[Quoted text hidden]

Matt McMillan <matt.s.mcmillan@gmail.com>
To: Benjamin Meyer <ben@kenaiwatershed.org>

Tue, Jun 9, 2026 at 3:43 PM



MatSuBorough_Lake_WQ.zip

Ben,

Here is a link to download the data. Let me know if you have any issues or questions.

The original data from MSB is in the MSBVLMP folder with individual folders and spreadsheets for each lake in the MASTER LAKE FILES folder.

I copied all the lake profile data from those to a csv named lake_profiles_all_46.csv which contains depth, temp, specific conductivity, DO, an pH as well as some other categorical data. The other table in this zipped folder is the turbidity (secchi disc), chlorophyll-a and phosphorus data, which I believe was taken at 2' or 2 meters. I only ever compiled the trophic state index data for 17 lakes, I can't remember why, I think it had to do with having a big enough sample size.

The other csvs are summary statistics for the 46 lakes, including some stable isotope data that was collected in 2015 or 2016.

If there is a particular part of my thesis you want more info on I can share R scripts or plots. I am sure if we had a conversation you would jog my memory about some aspects of it, so don't hesitate to reach out.

Hope this helps!

Matt

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Benjamin Meyer <ben@kenaiwatershed.org>
To: Matt McMillan <matt.s.mcmillan@gmail.com>

Tue, Jun 9, 2026 at 3:59 PM

Yahoo! Thanks Matt. I knew it was all out there somewhere.

I do not have specific plans for right now, except to get all the data poised in a new GitHub [repo](#) for a future effort. If you're a GitHub user also, I can add you as a contributor to the repo and you can add whatever other files you wish to share whenever you're able. Existing R scripts would be great.

Big picture, I want to use all this existing hard work to make the case for the value of continued monitoring of some sort. Maybe not in the same form as the MSVLMP, but something. A wise person once told me, "Two things you should never do are: a) start a long term monitoring project or b) end a long term monitoring project." A full effort once every five years, for example, might be enough to still be useful. We'll see what the data suggests. I collected some of the Lake Lucille data myself.

Thanks a million. Take care and be in touch if you ever need a favor in my neck of the woods. Cheers --

Ben

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Matt McMillan <matt.s.mcmillan@gmail.com>
To: Benjamin Meyer <ben@kenaiwatershed.org>

Wed, Jun 10, 2026 at 7:34 AM

Glad someone is taking an interest and has time to organize the data. Please add me to the repo, You can add both of these accounts <https://github.com/matt-s-mcmillan> or my RDI account <https://github.com/matthew-mcmillan>

I'll see if I can add some of my scripts as I have time and make sense of them. It is certainly a blast from the past, can't believe it has been over 10 years.

I can't remember if I talked about it much in my thesis, but I always thought it might be useful to implement a remote

sensing aspect to the monitoring, both ice-off/on date tracking and productivity using NIR bands or band ratios to quantify chlorophyll-a, or at least a proxy. But there really is no substitute for in-situ monitoring. And one of the most interesting things I discovered about a week before I presented my thesis was the increase in wind events that carry silt from the Matanuska braid plain across the valley. If I had another year I would have tried to understand how that affects lake water quality. My suspicion is that it had the biggest impact on the increases in conductivity in certain lakes. Those wind events in May are gnarly. I have a picture from the top of the Butte looking down on the braid plain with plumes of silt rising hundreds of feet in the air.

Lastly, there was related lake core sampling done by Lesleigh Anderson and Bruce Phinney that I helped with for Neklason Lake, Finger, and a small marl lake they named April Fool's Lake near Big Lake, because we cored it on April 1. I'll see if I can track down the papers on those, but it certainly adds some context to current conditions to show the geologic record with sediment deposition rates, nutrient inputs, and proxies for productivity going back ~10,000 years.

Matt

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